

AFL Assistant — Executive Report

Week 3 Day 5 Capstone

1. Product Goal

The AFL Assistant is a domain-locked chat and prediction system designed to provide useful, grounded assistance for Australian rules football. Its goal is to answer AFL factual and statistical questions, retrieve player and match information, and provide model-based predictions while preventing off-topic and prompt-injection requests.

The system combines three core capabilities:

- **Chat:** Handles natural-language AFL questions and maintains conversation context using conversation IDs.
- **Retrieval:** Provides factual and statistical information from the structured AFL data/retrieval layer.
- **Prediction:** Uses trained Week 3 prediction pipelines for match-winner and top-player predictions.

Predictions are presented with a shared disclaimer: **“Predicted probability, not a certainty. Actual results can differ.”**

The system is exposed through a FastAPI /chat endpoint and includes a minimal browser interface for demonstration.

2. Architecture

The assistant follows a controlled **LangGraph-oriented architecture** in which the user's request is routed according to its intent rather than allowing unrestricted tool usage.

High-level flow:

User → Scope/Safety Gate → Intent Routing → Chat / Retrieval / Prediction → Response → Logging

For an AFL factual or statistical question, the request is routed to the retrieval layer. Prediction questions are routed to the appropriate trained prediction pipeline. Off-topic and prompt-injection requests are blocked before tools are executed.

The system also includes:

- Conversation IDs for multi-turn interactions.
- Bounded tool execution and structured error handling.
- Prompt-injection detection.

- Off-topic scope protection.
- Structured JSON logging.
- Prediction metadata in API responses.
- Latency and estimated token-usage metadata.
- A /health endpoint for deployment monitoring.

The production wiring checks demonstrate that prediction requests can call the Day 2 prediction artifacts and factual/statistical requests can use the Day 3 retrieval adapter rather than relying only on demo callbacks.

3. Evaluation Results

The assistant was evaluated using **28 deterministic test cases** covering factual questions, prediction sanity checks, scope guardrails, prompt-injection attempts, and multi-turn conversation behavior.

The deterministic evaluation achieved:

Overall pass rate: 100% — 28/28 cases passed.

The evaluation included seven conversation cases covering follow-ups such as “What about that team?”, corrections such as “Actually I meant Geelong,” and player-stat follow-ups.

Prediction sanity tests also verified that stronger teams receive higher predicted probabilities than weaker teams in the designed test scenarios, and every prediction response includes the required uncertainty disclaimer.

For a real model-performance benchmark, the trained match-winner model was compared with a simple ladder-position/form proxy on the same chronological holdout:

Benchmark	Accuracy
Trained match-winner model	61.4%
Ladder-form proxy	53.2%
Difference	+8.2 percentage points

The benchmark contained **1,829 holdout matches** beginning in the 2017 holdout period.

The 100% deterministic result should therefore be interpreted as a **CI/safety/workflow benchmark**, not as proof that the underlying language model will achieve 100% real-world factual accuracy.

4. Known Limitations

Data recency:

The assistant depends on the available AFL datasets and retrieval artifacts. If new match results or player statistics are not ingested, responses can become outdated. The monitoring plan therefore checks data freshness and blocks model promotion when data-quality validation fails.

Model accuracy ceiling:

The match-winner model achieved 61.4% accuracy on the chronological holdout. This demonstrates useful predictive signal but also shows that match outcomes remain difficult to predict. A model probability should not be interpreted as certainty.

Evaluation limitations:

The 28-case deterministic suite verifies routing, guardrails, and expected behavior, but it does not fully measure open-ended LLM factual quality. The notebook itself identifies replacing deterministic factual/conversation cases with actual Day 3 retrieval traces as an important improvement.

Guardrail edge cases:

Prompt-injection and off-topic protection currently relies substantially on explicit detection patterns and scope checks. Unusual or previously unseen attempts may require additional testing and guardrail rules.

API validation:

The FastAPI service and conversation flow have been smoke-tested, but the notebook's demonstrated checks are primarily in-process rather than a full externally deployed production service test.

5. Recommended Next Steps

1. **Connect all factual and conversation evaluation cases to real Day 3 retrieval traces** instead of deterministic demo answers.
2. **Expand evaluation** with sampled human review and additional adversarial/prompt-injection cases.
3. **Improve data freshness** by automating ingestion of completed match results and player statistics.
4. **Monitor prediction quality** using accuracy, Brier score, calibration, and top-player hit rate after every completed round.
5. **Run the weekly model-refresh process** only after data-quality validation and comparison against the incumbent model and ladder baseline.

6. **Strengthen production deployment** with real HTTP/Uvicorn testing, provider-level token usage, persistent conversation storage, and stronger operational monitoring.

The monitoring plan already defines thresholds for latency, tool reliability, scope leakage, abuse pressure, prediction drift, and data freshness, with a weekly refresh and rollback process.

Conclusion

The AFL Assistant demonstrates the core components of a domain-specific production-oriented assistant: controlled conversational routing, retrieval, prediction, safety guardrails, API exposure, evaluation, and monitoring.

The current results show a **100% pass rate on the deterministic 28-case suite** and a **61.4% match-winner accuracy compared with 53.2% for the ladder-form benchmark**. The next stage should focus on replacing deterministic fixtures with fully integrated retrieval/conversation evaluation and improving freshness, observability, and real-world quality measurement.